

PROJECT REFERENCE · BILLFLOW

SAP-Style Automatic Billing Platform

Development plan with activity-level day estimates

Location: D:\Cursor\Billings\BillFlow

Status: Plan only — implementation not started

Last updated: 20 August 2026

Customer request in → validated, priced, invoiced, sent out → or routed to exception queue with a clear reason.

Item	Value
Project name	BillFlow
Folder	D:\Cursor\Billings\BillFlow
Team for estimate	1 full-time developer (backend + basic UI)
Working day	1 day = 6–7 focused hours
Calendar total	88 working days (~4.5 months) for Phases 0–5
2-developer option	About 50–55 working days (~2.5–3 months)
Buffer	+15% recommended for unknowns (not in totals)

1. Product summary

BillFlow (also referred to as **Billings**) is an orchestrated billing platform. A **customer request** is validated against **master data and business rules**, **priced and taxed automatically**, converted into **billing documents** (invoices, credit memos), and **delivered to the customer**. Exceptions go to staff. **LLM is optional** and limited to unstructured intake and customer messaging — not pricing or tax math.

Terminology

Term	Meaning in BillFlow
Orchestration	Workflow engine: plan steps, pass evidence, retries, audit
Module / worker	Single capability: intake, pricing, invoice PDF, etc.
Agent	Optional LLM + tools (email parse, exception reply drafts)
Mode	Runtime behaviour: AUTO, DRAFT_ONLY, LLM_ASSISTED_INTAKE

2. Scope reference (SAP-like areas)

Area	In scope (phased)
Customer / product master	Yes — Phase 0–1
Pricing & contracts	Yes — Phase 1–2
Tax determination	Yes — Phase 2 (rules); Phase 5 (external API optional)
Billing documents (invoice, credit memo)	Yes — Phase 1–4

Area	In scope (phased)
Output (PDF, email, portal)	Yes — Phase 2–3
Accounts receivable & dunning	Partial — Phase 4–5
GL / payment gateway / EDI	Phase 5

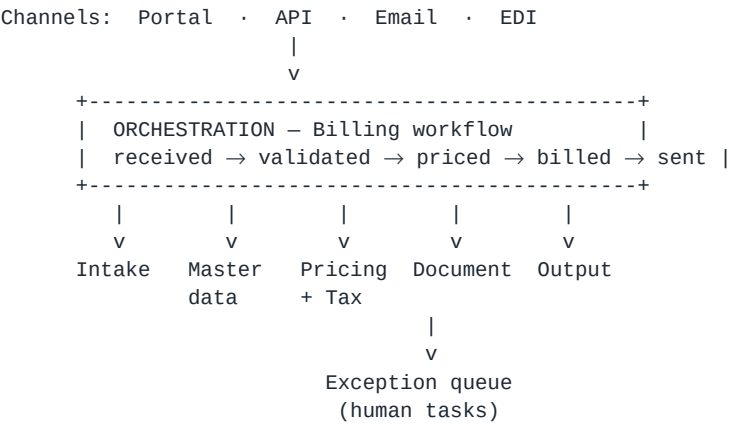
3. Assumptions for estimates

Assumption	Value
Team size	1 full-time developer (backend + basic UI)
Working day	1 day = 6–7 focused hours
Tech stack	Python or Node API, PostgreSQL/SQL Server, workflow layer
LLM	Optional; not on critical pricing/tax path
SAP integration	Out of scope for v1; design hooks only
Buffer	+15% recommended for unknowns (not included in table totals)

Calendar total (single developer): about 88 working days (~4.5 months) for Phases 0–5.

4. Architecture overview

Channels (portal, API, email, EDI) feed one orchestrator. The orchestrator moves a case through received → validated → priced → billed → sent. Specialist modules do one job each. Failures go to an exception queue.



Core modules

Module	Responsibility	LLM?
Intake	Accept and normalise requests	Optional (email)
Master data	Customer, product, terms lookup	No
Pricing	Price lists, discounts, contracts	No
Tax	Jurisdiction rules	No
Credit / risk	Limit checks, blocks	No
Billing document	Invoice / credit memo lifecycle	No
Output	PDF, email, portal	No
AR / payments	Open items, allocation	No

Module	Responsibility	LLM?
Exception	Task queue, reason codes	Optional (draft replies)

5. End-to-end automatic flow

- Customer submits request (API / portal / email)
- Intake validates schema and idempotency
- Master data resolves customer and products
- Credit module approves or holds
- Pricing module calculates line amounts
- Tax module applies rates
- Billing document module creates POSTED invoice
- Output module generates PDF and sends email
- On any failure → Exception module and audit log

6. Development phases — summary

Phase	Name	Duration	Cumulative
0	Design and data model	8 days	8
1	Core billing (draft path)	18 days	26
2	Full automatic processing	14 days	40
3	Customer channels and portal	12 days	52
4	Extended billing types and AR	18 days	70
5	Enterprise integration	18 days	88

7. Phase 0 — Design and data model (8 days)

#	Activity	Days	Deliverable
0.1	Requirements workshop: request types, statuses, roles	1	Scope doc
0.2	ERD: customers, products, contracts, requests, invoices	2	ERD diagram + table list
0.3	Billing request / invoice JSON schemas (OpenAPI draft)	1	schemas/ spec
0.4	Workflow state machine design (statuses, transitions)	1	State diagram
0.5	Rule catalog v1 (validation, pricing, credit, auto-post)	1	Rule matrix
0.6	Non-functional: idempotency, audit, security, numbering	1	NFR checklist
0.7	Seed data plan (sample customers, products, prices)	1	DATA.md equivalent

Exit criteria: Signed-off ERD, API contract for “submit request”, workflow states, rule catalog.

8. Phase 1 — Core billing, draft path (18 days)

#	Activity	Days	Deliverable
1.1	Project scaffold: repo, DB migrations, config	2	Runnable skeleton
1.2	Customer master CRUD + lookup APIs	2	customers module
1.3	Product catalog CRUD + lookup APIs	2	products module
1.4	Price list storage + simple base-price lookup	2	pricing module (basic)
1.5	Intake module: POST billing request, validation	2	Intake API
1.6	Orchestration skeleton: state store + step runner	3	Workflow engine v1
1.7	Billing document module: DRAFT invoice header/lines	3	Draft invoice API
1.8	Unit tests for intake + master + draft invoice	1	Test suite v1
1.9	Seed script: 10 customers, 20 products, 3 price lists	1	init_db script

Exit criteria: API accepts request → returns a draft invoice with priced lines (fixed price list, no tax/credit yet).

9. Phase 2 — Full automatic processing (14 days)

#	Activity	Days	Deliverable
2.1	Contract / customer-specific pricing rules	2	Pricing rules engine
2.2	Discount and surcharge rules (volume, promo codes)	2	Extended pricing
2.3	Tax module: region → rate, line tax amounts	2	Tax calculation
2.4	Credit module: limit, open AR, block flags	2	Credit decision API
2.5	Invoice number range (sequential, concurrency-safe)	1	Number service
2.6	Auto-post rules: when DRAFT → POSTED	1	Rule engine hook
2.7	Output module: PDF template + file storage	2	Invoice PDF
2.8	Email delivery (SMTP / SendGrid)	1	Send invoice job
2.9	Idempotency: duplicate request → same invoice	1	Idempotency keys

Exit criteria: Happy-path request fully automated: validate → price → tax → post → PDF → email, no manual step.

10. Phase 3 — Customer channels and portal (12 days)

#	Activity	Days	Deliverable
3.1	Public REST API hardening (auth, rate limits)	2	Secured API
3.2	Customer portal: submit order, view invoices	4	Simple web UI
3.3	Internal clerk UI: search cases, view audit trail	3	Ops UI
3.4	Exception queue UI: list, assign, approve/reject	2	Exception module UI
3.5	Webhook / callback on invoice posted	1	Event notifications

Exit criteria: Customer can submit via portal/API and receive invoice; clerk can see exceptions.

11. Phase 4 — Extended billing types and AR (18 days)

#	Activity	Days	Deliverable
4.1	Credit memo flow (return authorization link)	3	Credit memo module
4.2	Debit memo / invoice correction (delta billing)	3	Correction module
4.3	Recurring billing scheduler (subscriptions)	4	Scheduler + jobs
4.4	AR open items: post invoice to receivables ledger	2	AR sub-ledger
4.5	Payment recording and allocation to invoices	3	Payment module
4.6	Dunning: overdue rules + reminder letters	2	Dunning job
4.7	Reporting: billing run summary, exception KPIs	1	Reports API

Exit criteria: Credit/recurring paths work; basic AR and dunning operational.

12. Phase 5 — Enterprise integration (18 days)

#	Activity	Days	Deliverable
5.1	GL export (CSV / journal entry format)	3	FI export
5.2	Payment gateway integration (Stripe/Razorpay/etc.)	4	Online pay link
5.3	EDI / flat-file outbound invoices (optional format)	4	EDI adapter
5.4	External tax API adapter (Avalara-style hook)	3	Tax provider plug-in
5.5	Optional LLM intake: parse email → structured request	3	LLM agent module
5.6	Observability: structured logs, metrics, case tracing	1	Dashboards

Exit criteria: Finance export works; optional channels and LLM intake documented and gated by mode.

13. Orchestration implementation options

Option	Dev effort	Best for
Custom state machine	Included in Phase 1 (3 days)	Full control, SAP-like statuses
LangGraph ReWOO	+2 days on top of 1.6	Fixed pipelines with evidence #E1...#En
Temporal / Camunda	+5 days setup	Long-running, human tasks, retries

Recommendation for v1: custom state machine in Phase 1; add ReWOO-style evidence logging for audit (similar to the Manufacturer project).

14. Automatic processing rule catalog (v1)

Rule ID	Condition	Action	Module
R-001	Required fields missing	Reject + reason code	Intake
R-002	Unknown customer	Exception: master data	Master data
R-003	Customer blocked	Reject	Master data
R-004	Unknown product	Exception: catalog	Master data
R-005	No price found	Exception: pricing	Pricing
R-006	Credit exceeded	Hold → exception queue	Credit
R-007	All checks pass	Auto-post invoice	Orchestration
R-008	Duplicate idempotency key	Return existing invoice	Intake
R-009	Tax jurisdiction unknown	Exception: tax setup	Tax
R-010	Post success	Generate PDF + email	Output

15. Data model (minimum entities)

Entity	Purpose
customers	Bill-to, terms, credit limit, block flag
products	SKU, UoM, tax classification
price_lists / contract_prices	Effective-dated pricing
billing_requests	Incoming customer requests
billing_request_lines	Line items on request
invoices	Posted billing documents
invoice_lines	Priced and taxed lines
invoice_numbers	Sequence control
payments / allocations	AR (Phase 4)
workflow_events	Audit trail per case
exceptions	Human tasks

Phase 0 activity 0.2 produces the full ERD. Seed data covers about 10 customers and about 50 invoice scenarios.

16. API sketch (Phase 1 target)

Request

POST /api/v1/billing-requests
 Idempotency-Key: {uuid}
 Content-Type: application/json

```
{
  "customer_id": "C-10001",
  "po_reference": "PO-7788",
  "currency": "USD",
  "lines": [
    { "product_id": "SKU-001", "quantity": 10, "unit": "EA" }
  ]
}
```

Response — 201 Created

```
{
  "case_id": "BR-2026-00042",
  "status": "POSTED",
  "invoice_id": "INV-2026-001104",
  "pdf_url": "/artifacts/invoices/INV-2026-001104.pdf",
  "total_amount": 1250.00,
  "tax_amount": 125.00
}
```

17. LLM usage policy (modes)

Mode	LLM used for	Phases
AUTO	Nothing on the money path	1–4
DRAFT_ONLY	No post/send; preview only	2+
LLM_ASSISTED_INTAKE	Email → JSON request	5
LLM_EXCEPTION_COMMS	Draft clerk/customer emails	5 (optional)

Never use LLM for: tax rates, invoice totals, credit decisions, or number assignment.

18. Milestones and demo schedule

Milestone	Target (day)	Demo
M0 — Design complete	8	Walkthrough ERD + API
M1 — Draft invoice	26	Postman: request → draft PDF data
M2 — Auto invoice emailed	40	End-to-end happy path
M3 — Portal live	52	Customer submits order in UI
M4 — Credit memo + recurring	70	Two billing types
M5 — GL export + pay link	88	Finance handoff package

19. Risks and mitigations

Risk	Impact	Mitigation	Extra days
Pricing rules more complex than assumed	High	Phase 1 fixed price only; rules engine in 2.1	+3–5
Tax compliance	High	Start with static tables; external API in Phase 5	+0 (phased)
Concurrency on invoice numbers	Medium	DB sequence + locking in 2.5	Included
Unstructured email intake	Medium	Defer LLM to Phase 5; API first	Included
Scope creep ("full SAP")	High	Strict phase gates per this document	—

20. Suggested team parallelization (if 2 developers)

Track A (Core)	Track B (UI / integration)
Phases 0–2 modules	Portal mockups from day 10
Orchestration + pricing	Phase 3 UI
Phase 4 AR	Phase 5 GL export

With 2 developers: calendar time ≈ 50–55 working days (~2.5–3 months).

21. Relationship to Manufacturer project

Manufacturer	BillFlow
Analytic question → SQL → chart → PPT	Commercial request → price → invoice → PDF
LLM in presentation only	LLM in intake/comms only (optional)
ReWOO evidence #E1...#E3	Case evidence per workflow step
SQLite seed for demos	Master + transactional seed for billing scenarios

Patterns reused: orchestration folder, module per capability, seed data before agents/LLM, and modes for when AI is on.

22. Next steps (when you approve build)

- Confirm Phase 0 scope (which request types in v1).

- Choose database (PostgreSQL recommended) and API framework.
- Run Phase 0 activities 0.1–0.7 (8 days).
- Implement Phase 1 scaffold at D:\Cursor\Billings\BillFlow (separate from Manufacturer).

23. Document history

Version	Date	Author	Notes
1.0	20 Aug 2026	Plan	Initial development plan with day estimates
1.1	20 Aug 2026	Plan	PDF reference edition for project understanding